

MATE 6551: Tarea 2

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Problem 0

If $f : X \rightarrow Y$ is nullhomotopic and if $g : Y \rightarrow Z$ is continuous, then $g \circ f$ is nullhomotopic.

Proof:

Suppose that $f : X \rightarrow Y$ is nullhomotopic and $g : Y \rightarrow Z$ is continuous. Then there exists a homotopy $f \simeq k$ for some constant function $k : X \rightarrow Y$. Let $F : X \times I \rightarrow Y$ be such a homotopy. Then define $H : X \times I \rightarrow Z$ by $H(x, t) = g(F(x, t))$. Note that H is a composition of continuous functions and is, therefore, continuous. Moreover, $H(x, 0) = g(F(x, 0)) = g(f(x)) = (g \circ f)(x)$, and $H(x, 1) = g(F(x, 1)) = g(k(x)) = (g \circ k)(x)$. But k is constant, so $g \circ k$ must be constant as well.

$\therefore g \circ f$ is nullhomotopic.

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Problem 1

Prove that $X \mapsto CX$ defines a functor $\mathbf{Top} \rightarrow \mathbf{Top}$.

Proof:

The cone over a space X is $CX = \frac{X \times I}{X \times \{1\}}$. This defines the behavior of C on objects, we will define its behavior on morphisms.

Let $f : X \rightarrow Y$ be a continuous map, then define $Cf : CX \rightarrow CY$ by $Cf[(x, t)] = [(f(x), t)]$. To show that this map is well-defined, suppose that we have $[(x, t)] = [(x', t')]$ in CX , then $t = t'$ by construction of CX . If $t = t' = 1$, then:

$$Cf[(x, t)] = [(f(x), t)] = [(f(x), 1)] = [(f(x'), 1)] = [(f(x'), t')] = Cf[(x', t')] \quad (1)$$

If not, then $x = x'$, again by construction of CX . Thus,

$$Cf[(x, t)] = [(f(x), t)] = [(f(x'), t')] = Cf[(x', t')] \quad (2)$$

Therefore, Cf is well-defined.

Let $\pi : Y \times I \rightarrow CY$ denote the quotient application, and $c : CX \rightarrow X \times I$ be a continuous choice function picking out a representative for every class. Then $Cf = \pi \circ (f \times 1_I) \circ c$, a composition of continuous functions. Therefore Cf is continuous.

We now show that C preserves compositions. Let $g : Y \rightarrow Z$ be a continuous map. Then:

$$\begin{aligned} C(g \circ f)[(x, t)] &= [((g \circ f)(x), t)] = [(g(f(x)), t)] \\ &= Cg[(f(x), t)] = Cg[Cf[(x, t)]] = (Cg \circ Cf)[(x, t)] \end{aligned} \quad (3)$$

Furthermore, it preserves identities:

$$C1_X[(x, t)] = [(1_X(x), t)] = [(x, t)] = 1_{CX}[(x, t)] \quad (4)$$

$\therefore C$ is a functor.

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Problem 2

If X and Y are path connected, then $X \times Y$ is path connected.

Proof:

Suppose that X and Y are path connected, and take $(x, y), (x', y') \in X \times Y$. Then $x, x' \in X$ and $y, y' \in Y$, but X and Y path connected $\implies \exists f : I \rightarrow X, g : I \rightarrow Y$ paths from x to x' and from y to y' respectively. Now define $\varphi : I \rightarrow X \times Y$ to be $\varphi = f \times g$, that is, $\varphi(t) = (f(t), g(t))$. This map is clearly continuous, and satisfies:

$$\varphi(0) = (f(0), g(0)) = (x, y) \quad \text{and} \quad \varphi(1) = (f(1), g(1)) = (x', y') \quad (5)$$

Thus φ is a path from (x, y) to (x', y') .

$\therefore X \times Y$ is path connected.

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Problem 3

If $f : X \rightarrow Y$ is continuous and X is path connected, then $f(X)$ is path connected.

Proof:

Suppose that $f : X \rightarrow Y$ is continuous and that X is path connected. Take $f(x), f(x') \in f(X)$. Then $x, x' \in X$, but X path connected $\implies \exists g : I \rightarrow X$ a path from x to x' in X . Defining $\varphi : I \rightarrow f(X)$ as the composite $\varphi = \tilde{f} \circ g$, where $\tilde{f} : X \rightarrow f(X)$ denotes the co-restriction of f to its image, yields a path from $f(x)$ to $f(x')$. Indeed, φ is clearly continuous, and:

$$\varphi(0) = \tilde{f}(g(0)) = \tilde{f}(x) = f(x), \quad \varphi(1) = \tilde{f}(g(1)) = \tilde{f}(x') = f(x') \quad (6)$$

$\therefore f(X)$ is path connected.

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Problem 4

The path components of a space X are maximal path connected subspaces; moreover, every path connected subset of X is contained in a unique path component of X .

Proof:

We recall that, by definition, a path component of a space X is an equivalence class of points of X under the equivalence relation given by $x \sim y \iff$ there exists a path from x to y .

Let X be a topological space and $[x] \subseteq X$ a path component. Suppose that $Y \subseteq X$ is a path connected subspace of X containing $[x]$, and take $y \in Y$. Since $[x] \subseteq Y$, we know that there must be a path from x to y , but that implies that $x \sim y$ and thus $y \in [x]$. Therefore, $Y = [x]$.

$\therefore [x]$ is maximal with respect to path connected subspaces of X .

Let $Z \subseteq X$ be a path connected subspace of X , and take $z_0 \in Z$. Path connectedness implies that, for every $z \in Z$, there exists a path from z to z_0 . That is,

$$z \sim z_0 \quad \forall z \in Z \implies z \in [z_0] \quad \forall z \in Z \implies Z \subseteq [z_0] \quad (7)$$

Therefore, Z is contained in a path component. For uniqueness, suppose that $[z_0], [z_1]$ are path components with $Z \cap [z_0] \neq \emptyset$ and $Z \cap [z_1] \neq \emptyset$. Then $\exists p_0 \in Z \cap [z_0]$, and $p_1 \in Z \cap [z_1]$. But Z is path connected, so there must be some path from p_0 to p_1 . That is, $p_0 \sim p_1$, but we had $p_0 \in [z_0]$ and $p_1 \in [z_1]$. Then by transitivity, $z_0 \sim p_0 \sim p_1 \sim z_1 \implies [z_0] = [z_1]$.

$\therefore Z$ is contained in a unique path component of X .

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